A DIGITAL FOOD MANAGEMENT SYSTEM FOR COLLEGE CANTEEN

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINAR

EXTERNAL EXAMINAR

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CHAPTER – 1

INTRODUCTION

In the modern digital era, technology plays a crucial role in improving everyday services and systems. College canteens, which serve a large number of students daily, often face challenges in managing food orders efficiently. During peak hours, students experience long waiting times and overcrowding, which reduces convenience and productivity. Hence, there is a need for a smart and efficient solution to streamline canteen operations.

Traditional canteen systems are mostly manual or partially digital, where orders are placed directly at the counter. This approach leads to delays, human errors, and difficulty in handling large volumes of orders. Moreover, there is limited flexibility for modifying or canceling orders once placed. These limitations highlight the need for a fully digital system that can improve accuracy and reduce operational issues.

The Digital Food Management System is designed to address these challenges by introducing an online food ordering platform. It allows users to place orders in advance, receive real-time updates, and collect food using a token-based system. This approach significantly reduces queue formation and ensures better time management for both students and canteen staff. It also enhances the overall user experience through convenience and speed.

Additionally, the system helps in reducing food wastage by ensuring timely collection of orders and proper tracking. Features such as order notifications, token expiry alerts, and flexible order management make the system more efficient and user-friendly. Overall, this project aims to create a modern, organized, and sustainable canteen environment through the integration of digital technology

1.1 ABOUT THE PROJECT

The Digital Food Management System for College Canteen is an innovative solution designed to improve the efficiency of food ordering and distribution in college environments. It aims to replace traditional manual ordering methods with a fully digital system that simplifies the process for both students and canteen staff. By integrating modern web technologies, the system ensures faster service, reduced waiting time, and a more organized approach to canteen management.

The main objective of this project is to enable users to place food orders online through a user-friendly interface. Students can browse the menu, select items, and confirm their orders without standing in long queues. Once the order is placed, the system generates a unique digital token, which is used for food collection. This token-based approach helps in managing crowd flow effectively and ensures smooth and systematic order delivery.

The system also provides real-time updates on order status, allowing users to track their orders easily. Features such as order cancellation, modification, and notification alerts enhance user convenience and flexibility. On the management side, the system helps canteen staff handle multiple orders efficiently, reduces manual errors, and improves overall coordination in food preparation and delivery.

In addition, the project focuses on minimizing food wastage and improving resource management. By ensuring timely notifications and structured order handling, uncollected orders can be reduced significantly. The Digital Food Management System ultimately creates a smart, reliable, and sustainable canteen environment, enhancing user satisfaction and operational performance through the effective use of digital technology.

1.2 PROBLEM STATEMENT

In college canteens, managing food orders efficiently has become a significant challenge due to the increasing number of students depending on canteen services every day. During peak hours such as lunch breaks, students are required to stand in long queues to place and receive their orders, leading to overcrowding and loss of valuable time. The existing system is mostly manual or partially digital, which makes the ordering and billing process slow and inefficient. Manual handling of orders often results in errors such as incorrect billing, missing items, or confusion between orders. There is no proper real-time tracking system, making it difficult for students to know the status of their orders. This lack of transparency causes delays and dissatisfaction among users. Additionally, there are no effective notification systems to inform students when their orders are ready, leading to further delays in collection. As a result, many orders remain uncollected, contributing to unnecessary food wastage. The current system also lacks flexibility, as users cannot easily modify or cancel their orders once placed. Canteen staff face difficulties in managing large volumes of orders efficiently due to the absence of an organized system. This increases workload, reduces productivity, and affects the overall service quality. Moreover, the absence of proper data management makes it hard to analyze demand and plan resources effectively. These limitations clearly indicate the need for a smart, automated, and user-friendly solution. Therefore, the Digital Food Management System is proposed to overcome these issues by streamlining the ordering process, reducing queues, and improving overall efficiency.

1.3 OBJECTIVE

The main objective of this project is to develop a Digital Food Management System that simplifies and automates the food ordering process in college canteens. It aims to reduce long queues and waiting time by allowing users to place orders online through a user-friendly interface. The system focuses on providing real-time order tracking and a token-based collection method to ensure smooth and efficient service. It also helps in minimizing human errors in order processing and billing. Another key objective is to reduce food wastage by ensuring timely notifications and proper order management. Overall, the project enhances user convenience, improves operational efficiency, and creates a smart and organized canteen environment.

1.4 SCOPE AND LIMITATION OF THE STUDY

Scope:

The Digital Food Management System is designed to improve the efficiency of food ordering and service in college canteens by introducing a fully digital platform. It allows users to browse menus, place orders online, and make use of a token-based system for food collection. The system supports real-time order tracking, helping users stay informed about their order status. It can be implemented using web technologies, making it easily accessible through mobile phones and computers. The scope also includes features like order cancellation, notifications, and better queue management. It helps canteen staff manage large volumes of orders efficiently and reduces manual workload. Additionally, the system can be extended in the future to include online payment options and data analytics. Overall, the project aims to create a smart, scalable, and user-friendly solution for modern canteen management.

Limitations:

Despite its advantages, the Digital Food Management System has certain limitations that may affect its implementation. The system requires a stable internet connection for both users and canteen staff to function effectively. Users who are not familiar with digital platforms may find it difficult to adapt initially. The development and maintenance of the system may involve additional costs for software setup. Security concerns such as data privacy and unauthorized access need to be carefully managed. The system may also face technical issues like server downtime or software bugs. It depends heavily on user participation for successful implementation and smooth functioning. Additionally, integrating advanced features like online payments may require further development and testing. These limitations highlight the need for proper planning and continuous improvement of the system.

CHAPTER – 2

LITERATURE SURVEY

2.1 INTRODUCTION ABOUT LITERATURE SURVEY

The literature survey plays an important role in understanding the existing research and technologies related to digital food ordering and canteen management systems. It involves studying various previously developed systems, research papers, and applications that focus on improving food service efficiency. By analyzing these works, it becomes easier to identify the strengths and weaknesses of current solutions. This helps in gaining knowledge about different approaches and technologies used in similar projects.

Several studies have proposed online food ordering systems using web and mobile applications, along with technologies such as QR codes, cloud databases, and secure payment systems. These systems aim to reduce manual work, improve order accuracy, and enhance user convenience. Some advanced systems also include features like real-time order tracking, digital payments, and automated notifications. However, certain solutions depend heavily on specific hardware or require complex implementation, which may not be suitable for all environments.

The literature survey helps in identifying research gaps and areas for improvement in existing systems. Based on this analysis, the proposed Digital Food Management System focuses on providing a simple, cost-effective, and user-friendly solution tailored for college canteens. It combines essential features such as online ordering, token-based collection, and real-time updates to overcome the limitations of previous systems. Thus, the survey provides a strong foundation for developing an efficient and practical system.

2.2 REVIEW OF LITERATURE SURVEY

Various research studies have focused on developing digital food ordering systems to improve canteen efficiency and reduce manual work. Many systems use web and mobile applications to enable online ordering and enhance user convenience. Some approaches integrate technologies like QR codes and IoT devices to streamline the ordering process. Real-time order tracking and notification features are commonly implemented to improve transparency and reduce waiting time. Certain systems also include digital payment options for faster and cashless transactions. However, some solutions rely on complex hardware setups, increasing cost and maintenance requirements. Other studies highlight issues such as limited scalability, security concerns, and dependency on internet connectivity. Based on these findings, the proposed system aims to provide a simple, cost-effective, and efficient solution tailored for college canteens.

1. Online Food Ording System For College Canteens

Author: Rupali B. Kale, Ruchika K. Balwade, and Vipin B. Gawai

Year of Publication: 2020

This paper presents the design and development of a digital solution aimed at improving the efficiency of food services in a college campus. The study begins by identifying the major drawbacks of the traditional canteen system, such as long waiting lines, delays during peak hours, manual order handling, and frequent errors in order processing. To overcome these challenges, the authors propose an online platform that enables students and staff to view the menu, select items, place orders in advance, and complete payments through a web or mobile interface. The system includes features like real-time order tracking, order history, and automated billing, which help in reducing confusion and improving transparency. It also assists canteen administrators in managing orders systematically, maintaining records, and optimizing food preparation based on demand. The implementation of this system not only minimizes crowding and saves time but also enhances overall user satisfaction by providing a convenient, fast, and organized food-ordering experience within the campus.

2. Canteen Management Application

Author: Ganadhish Navelkar, Bidesh Chanda, Vaibhav Sharma, Atish Gracias, Basil Jose, and Valerie Menezes

Year of Publication: 2022

This reference discribes the working of conventional canteen systems and the challenges associated with them. It explains how food ordering and service are typically carried out manually, involving direct interaction between customers and staff, which often leads to inefficiencies such as long queues, delays in service, and miscommunication of orders. The study highlights issues like poor record maintenance, difficulty in managing peak-hour demand, and lack of proper coordination between order taking and food preparation. It also discusses how these limitations affect both customer satisfaction and overall productivity of the canteen. By analyzing these drawbacks, the paper emphasizes the need for modernization and suggests that adopting digital solutions, such as automated or online ordering systems, can significantly improve efficiency, accuracy, and service quality in canteen operations.

3.VIT Canteen

Author: Aneesh Surendra Oak, Prof. Gauri Kulkarni, Sanchitsai Baliram Nipanikar, Nishant Pravin Deore, Nitesh Rajendra Tandel, Om Lalit Kandpal, and Om Praful Poreddiwar

Year of Publication: 2023

The system focuses on the development and implementation of a smart canteen management system designed to improve food service operations within a college campus. The study highlights common issues in existing canteen systems, such as overcrowding, long waiting times, inefficient order handling, and lack of real-time communication between customers and staff. To address these challenges, the authors propose a digital solution that allows users to view menus, place orders online, and make payments through an integrated platform. The system may also include features like order tracking, notifications, and database management for maintaining records of transactions and customer preferences. Additionally, it helps canteen administrators manage inventory, monitor demand, and streamline the preparation process. Overall, the project demonstrates how the adoption of modern technology can enhance operational efficiency, reduce delays, and provide a more convenient and user-friendly experience for students and staff.

4.Canteen Automation System

Author: Mahiba, Rajasekar Dhanush, Santhosh Kumar, and Sharath Chandra

Year of publication: 2023

This study focuses on the design and implementation of a canteen automation system aimed at improving the overall efficiency of food ordering and management processes. The system replaces traditional manual methods with a digital solution that allows users to view menus, place orders, and make payments through an online platform. It helps in significantly reducing long queues, waiting time, and human errors commonly found in manual order handling.The proposed system includes features such as real-time order tracking, secure payment integration, and a user-friendly interface that enhances customer experience. It also maintains a centralized database to store transaction details, order history, and inventory information, which helps canteen staff manage operations more effectively. Additionally, the system ensures better communication between customers and kitchen staff, leading to improved order accuracy and faster service.Overall, the canteen automation system provides a modern, reliable, and efficient solution that benefits both customers and administrators by streamlining operations, improving service quality, and supporting better management of resources.

5.Canteen Food Ordering System and Management

Author: Bala Hariharan, Poongodi

Year of Publication: 2025

This study presents the design and development of a canteen food ordering and management system that aims to modernize traditional canteen operations using a digital platform. The system enables users to browse a detailed menu, place orders in advance, and make secure online payments, thereby reducing congestion, long queues, and waiting time in busy canteens. It incorporates features such as order scheduling, real-time order status tracking, and dynamic menu updates, which improve both customer convenience and service efficiency. Additionally, the system uses a well-structured database to store customer information, order history, payment details, and inventory data, allowing administrators to effectively monitor sales, manage stock, and generate reports. The study also highlights improved coordination between customers and kitchen staff, leading to greater order accuracy and faster delivery. Overall, the system offers a reliable, efficient, and user-friendly solution that enhances service quality and supports better management of canteen operation.

TABLE 2.1 INFERENCES FROM LITERATURE SURVEY

S. NO	Title of the paper	Author Details	Publication & Year	Remarks
1.	Online Food Ordering system for College canteen	Rupali B. kale et al.	Samriddhi Journal,2020	Automates Ordering; reduces manual work; hardware dependency (Raspberry Pi, QR, Wi-Fi)
2.	Canteen Management Application	Gandhi Haselkar et al.	IJRASET,2022	Enables remote ordering; improves convenience; limited security/ payment integration
3.	VIT Canteen Platform	Aneesh Surendra Et al	International Journal,2023	Smooth ordering and UPI payment; depends on user adoption and maintenance
4.	Canteen Automation system	Mahiba et al.	International Journal,2023	Secure database encryption; high admin control; complex and costly implementation
5.	Canteen Ordering System and Management	Bala Hariharan	IRE Journals, 2025	Efficient crowd management; real-time ordering; requires stable internet connection

CHAPTER – 3

SYSTEM REQUIREMENT SPECIFICATION

3.1 INTRODUCTION

The Digital Food Management System for College Canteen is designed to enhance the efficiency and convenience of food ordering and management in college environments. In the existing manual system, students often experience long queues, delays in service, and confusion during peak hours due to improper order handling. This results in time wastage, reduced efficiency, and sometimes food wastage due to uncollected orders.

To address these challenges, the proposed system introduces a fully digital platform that enables users to place food orders online through a web-based interface. The system generates a unique digital token for each order, which helps in smooth and organized food collection without crowding. It also provides real-time updates about order status, improving transparency between students and canteen staff.

The system reduces manual workload, minimizes errors in order processing, and improves overall service quality. It is designed to be user-friendly, reliable, and scalable, making it suitable for implementation in college canteens. By integrating modern web technologies and database management, the system ensures efficient handling of orders, better time management, and enhanced user satisfaction.

3.2 SOFTWARE DESCRIPTION

The Digital Food Management System is developed using modern web technologies that ensure flexibility and efficiency.

HTML (Hyper Text Markup Language):

Used to design the structure of web pages such as login page, menu display, and order interface.

CSS (Cascading Style Sheets):

Used for styling the application and improving layout, colors, and user interface.

JavaScript:

Used to add interactivity and handle dynamic operations like placing orders, generating tokens, and updating order status.

MongoDB:

A NoSQL database used to store user details, menu items, and order information efficiently.

CHAPTER – 4

PROJECT DESCRIPTION

4.1 EXISTING SYSTEM

The existing system used in most college canteens is primarily manual, where students and staff place their food orders directly at the counter. During peak hours, this leads to long queues and overcrowding, making it difficult to manage large numbers of customers efficiently. The process of taking orders, preparing food, and delivering it is handled manually, which increases the chances of errors. There is often confusion in identifying orders, especially when multiple orders are placed at the same time. The lack of a proper queue management system results in delays and dissatisfaction among users. Customers are required to wait for a long time, which affects their schedule and productivity. Additionally, there is no structured method to handle order prioritization. The absence of automation makes the overall system slow and less efficient. Manual record-keeping also makes it difficult to track daily transactions and performance. Overall, the existing system struggles to meet the demands of a large student population.

In some cases, canteens use partially digital systems, such as basic billing software or simple online ordering without full integration. However, these systems still require users to be physically present for order collection without any proper scheduling or token mechanism. There is no real-time order tracking, which causes uncertainty about order status. Notifications are either absent or very limited, leading to confusion and missed orders. This often results in food being wasted when customers fail to collect their orders on time. Furthermore, there is little to no flexibility for users to modify or cancel their orders after placing them. The system also lacks proper data analysis features, making it hard for management to predict demand and manage resources efficiently. Technical limitations and poor user interface design may also reduce user adoption. These partially digital systems fail to fully solve the problems of queue management and efficiency. Hence, the need for a fully automated and integrated digital solution becomes essential.

4.1.1 DRAWBACKS OF EXISTING SYSTEM

The existing canteen system has several drawbacks that affect both efficiency and user experience. One of the major issues is the presence of long queues during peak hours, which leads to overcrowding and delays. Students are forced to spend a lot of time waiting, which affects their academic schedule. The manual method of taking orders increases the chances of human errors such as incorrect entries and billing mistakes. There is also confusion in handling multiple orders at the same time. The absence of a proper queue management system makes the process unorganized. Customers often face inconvenience due to slow service and lack of coordination. Additionally, there is no proper tracking system to monitor order status. This results in uncertainty and dissatisfaction among users. Overall, the system is inefficient and unable to handle large volumes of customers effectively.

Another major drawback is the lack of flexibility and technological integration in the existing system. Users cannot easily modify or cancel their orders once placed, which leads to inconvenience. There are no notification systems to inform customers about order readiness, causing delays in collection. This often results in food wastage when orders are not picked up on time. The system also lacks real-time updates, making it difficult for users to plan accordingly. From the management side, there is no proper data storage or analysis for decision-making. This limits the ability to predict demand and manage resources efficiently. In partially digital systems, poor interface design and technical issues may reduce usability. The dependency on manual work increases workload and reduces productivity. Hence, these drawbacks highlight the need for a more advanced and automated solution.

4.2 PROPOSED SYSTEM

The proposed Digital Food Management System is designed to modernize and simplify the food ordering process in college canteens. It replaces the traditional manual system with a fully digital platform that allows users to order food conveniently. The system aims to reduce crowding and waiting time by enabling advance ordering. It provides a structured and organized method for handling large numbers of orders. By integrating technology, it improves efficiency and service quality. Overall, the system creates a better experience for both users and canteen staff.

The system includes a user-friendly interface where students can easily browse the menu and select food items. The menu is displayed digitally with clear details such as item names and availability. Users can add items to their cart and confirm orders with just a few clicks. This eliminates the need to stand in queues for placing orders. It also makes the process faster and more convenient. The interface is designed to be simple and easy to use for all users.

Once an order is placed, the system generates a unique digital token for each user. This token is used for collecting the food from the canteen. The token-based system helps in managing crowd flow effectively. It ensures that users receive their orders in an organized manner. This reduces confusion and avoids unnecessary delays. It also improves coordination between customers and canteen staff.

The system provides real-time updates on the status of each order. Users can track whether their order is being prepared or is ready for collection. Notifications are sent to inform users about important updates. This reduces uncertainty and helps users plan their time better. It also minimizes the chances of missing orders. Real-time tracking enhances transparency and reliability of the system.

Another important feature of the proposed system is flexibility in order management. Users can modify or cancel their orders within a certain time limit. This provides greater convenience and control over their orders. It also helps in reducing unnecessary food preparation. The system allows smooth handling of changes without confusion. This feature improves overall user satisfaction. It also supports efficient resource utilization.

The system also focuses on reducing food wastage in the canteen. By sending timely notifications, users are reminded to collect their orders. Uncollected orders can be tracked and managed properly. This helps in minimizing waste and improving sustainability. The system encourages better planning and usage of resources. It also supports a more responsible food management approach. Overall, it contributes to an eco-friendly environment.

From the management perspective, the system helps canteen staff handle orders more efficiently. It reduces manual work and minimizes errors in order processing. Staff can easily manage multiple orders through the digital interface. It improves coordination between order taking and food preparation. The system also stores data for future analysis and decision-making. This helps in better planning and improving service quality.

The proposed system is scalable and can be enhanced with additional features in the future. It can include online payment options and advanced analytics for demand prediction. The system can also be integrated with mobile applications for better accessibility. Continuous updates and improvements can make the system more efficient. It is designed to adapt to changing user needs and technological advancements. Overall, the proposed system provides a smart, efficient, and sustainable solution for modern canteen management.

4.2.1 ADVANTAGES OF PROPOSED SYSTEM

The proposed Digital Food Management System offers several benefits that improve both user experience and canteen operations. It helps in creating a more organized, efficient, and technology-driven environment for food service management.

1. Reduces long queues and overcrowding by enabling online food ordering.

2. Saves time for students and staff through faster and efficient service.

3. Provides real-time order tracking, improving transparency and convenience.

4. Token-based system ensures organized and smooth food collection.

5. Minimizes human errors in order taking and billing processes.

6. Allows order cancellation and modification, offering flexibility to users.

7. Sends notifications and alerts to avoid missing or delayed order collection.

8. Helps reduce food wastage by ensuring timely pickup of orders.

9. Improves overall efficiency and productivity of canteen operations.

10. Enhances user satisfaction with a simple and user-friendly interface.

CHAPTER – 5

IMPLEMENTATION AND RESULT

5.1 SYSTEM IMPLEMENTATION

Initially, users access the system by logging in through a secure interface. After successful login, they can view the digital menu, which displays available food items along with their prices. Users can select their desired items and place orders easily through the application.

Once an order is placed, the system generates a unique digital token for that order. This token acts as an identification number for food collection. The order details are stored in the database and are simultaneously sent to the canteen staff for preparation.

The staff prepares the food
The Digital Food Management System is implemented as a web-based application that simplifies the food ordering process in college canteens. The system connects students and canteen staff through a centralized platform, enabling smooth communication and efficient order handling.
based on incoming orders and updates the order status in the system. The user receives real-time updates such as “Order Placed,” “Preparing,” and “Ready.” When the order is ready, the user can collect the food by showing the digital token, avoiding long queues and confusion.

This implementation ensures accuracy, reduces manual work, and improves the overall efficiency of canteen operations.

5.2 SOFTWARE IMPLEMENTATION

The system is developed using HTML, CSS, JavaScript, and MongoDB to ensure smooth functionality and user interaction.

HTML is used to design the structure of the application, including login pages, menu display, and order sections.
CSS is used to enhance the visual appearance and layout of the application, making it user-friendly.
JavaScript is used to handle dynamic operations such as order placement, token generation, and real-time updates.
MongoDB is used as the database to store user data, menu details, and order records.

These technologies work together to provide a responsive and efficient system for managing food orders.

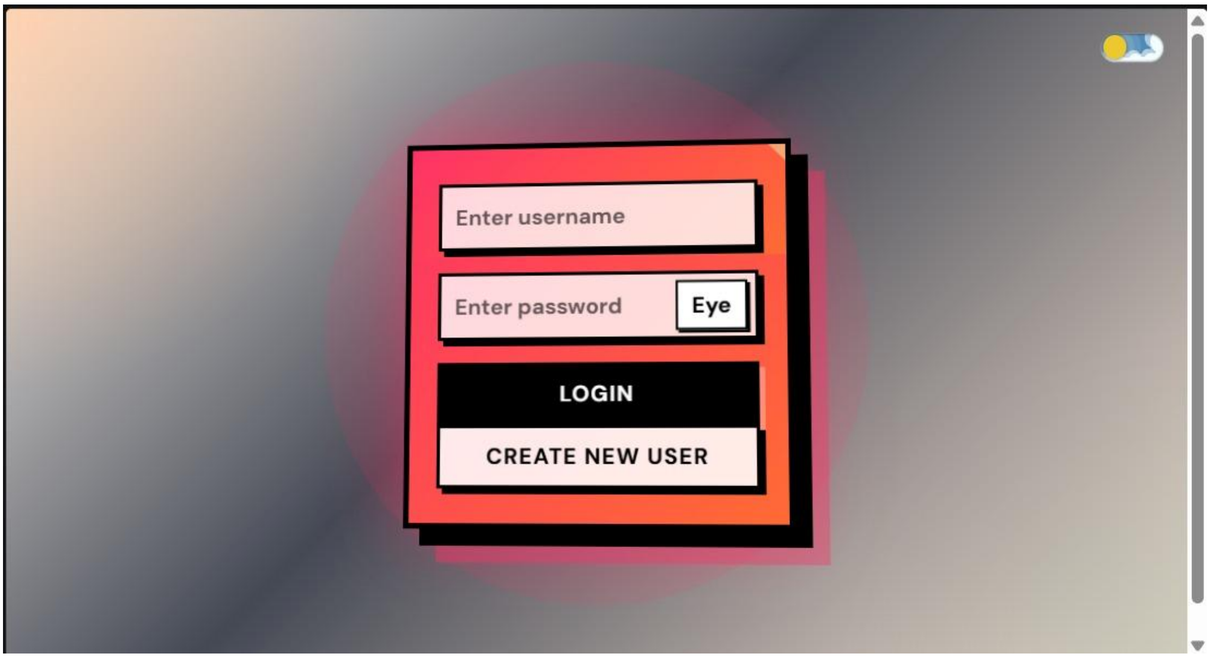
5.3 WORKFLOW

The Digital Food Management System operates through two main users: Students (Users) and Admin (Canteen Staff). The workflow for both is explained below:

5.3.1 Student Workflow

1.User Login / Registration:

The student logs into the system using valid credentials. If the user is new, they can register by providing necessary details. This ensures secure and personalized access.

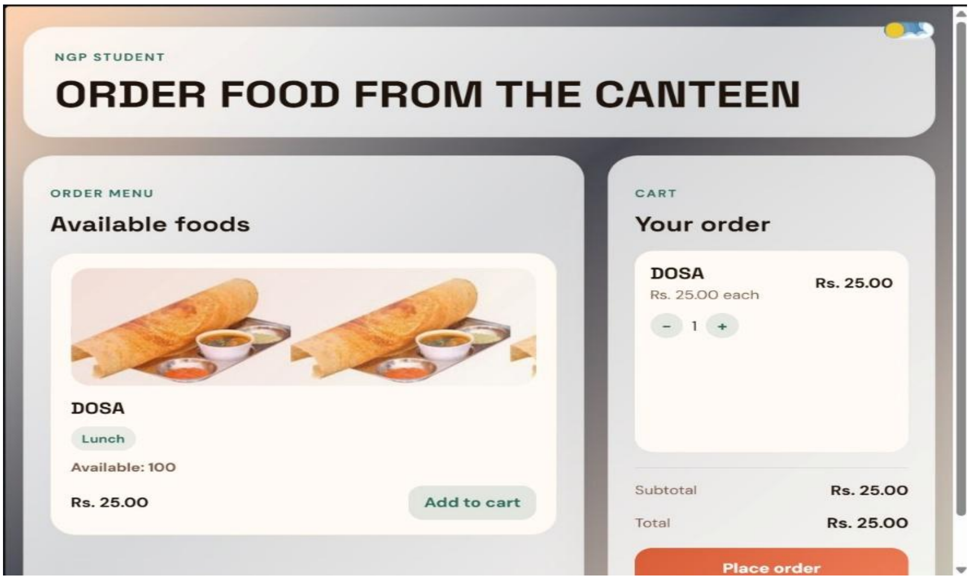


2.Viewing the Menu:

After logging in, the student can view the digital menu which displays all available food items along with their prices and details.

3.Selecting Food Items:

The student selects desired food items and adds them to the order. The system allows modification such as adding or removing items before final confirmation.

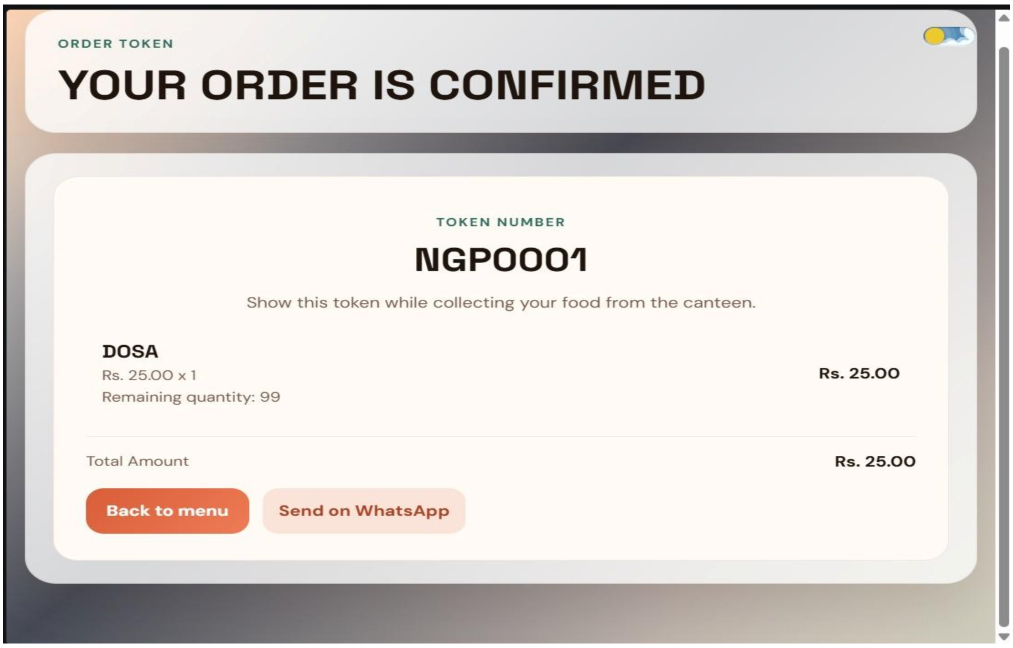


4.Placing the Order:

Once the selection is finalized, the student places the order. The system records the order details in the database.

5.Token Generation:

After successful order placement, a unique digital token is generated. This token is used as a reference for order tracking and food collection.

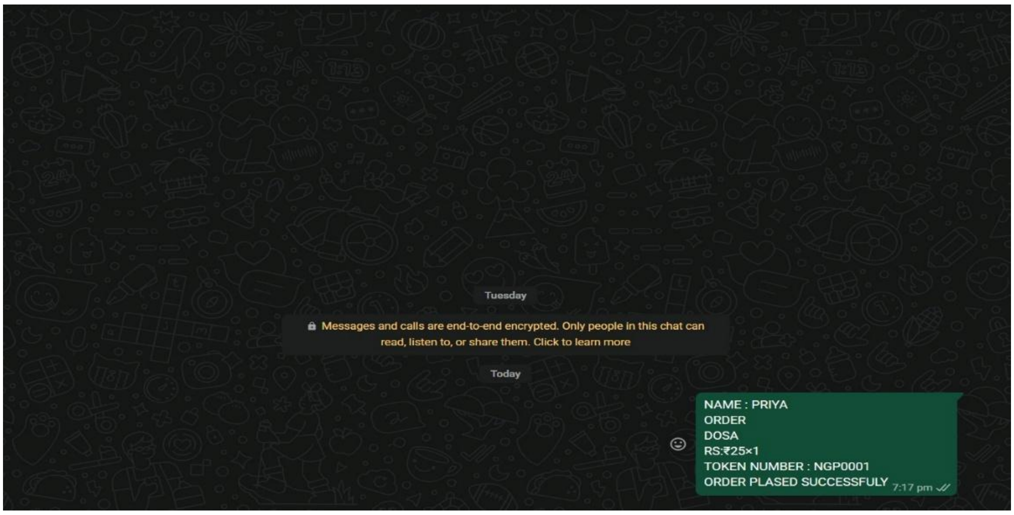


6.Order Tracking:

The student can view the real-time status of the order such as "Order Placed," "Preparing," and "Ready."

7.Notification:

When the order is ready, the student receives a notification through the system.



8.Food Collection:

The student goes to the collection counter and shows the digital token. The staff verifies the token and hands over the food.

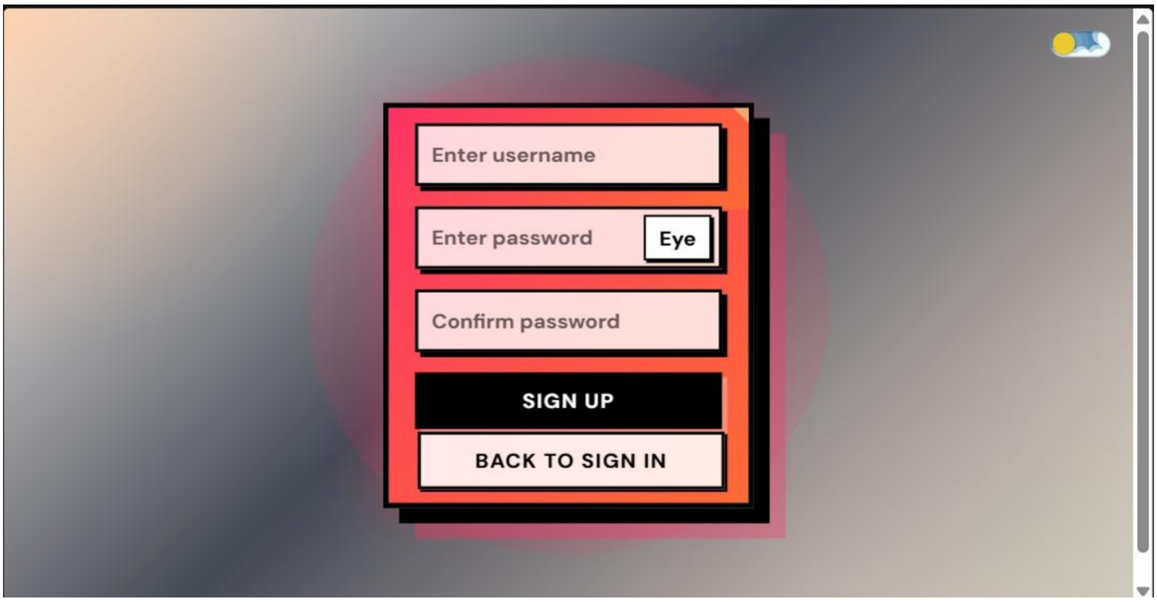
9.Order Completion:

After collecting the food, the order is marked as completed in the system.

5.3.2 Admin (Canteen Staff) Workflow

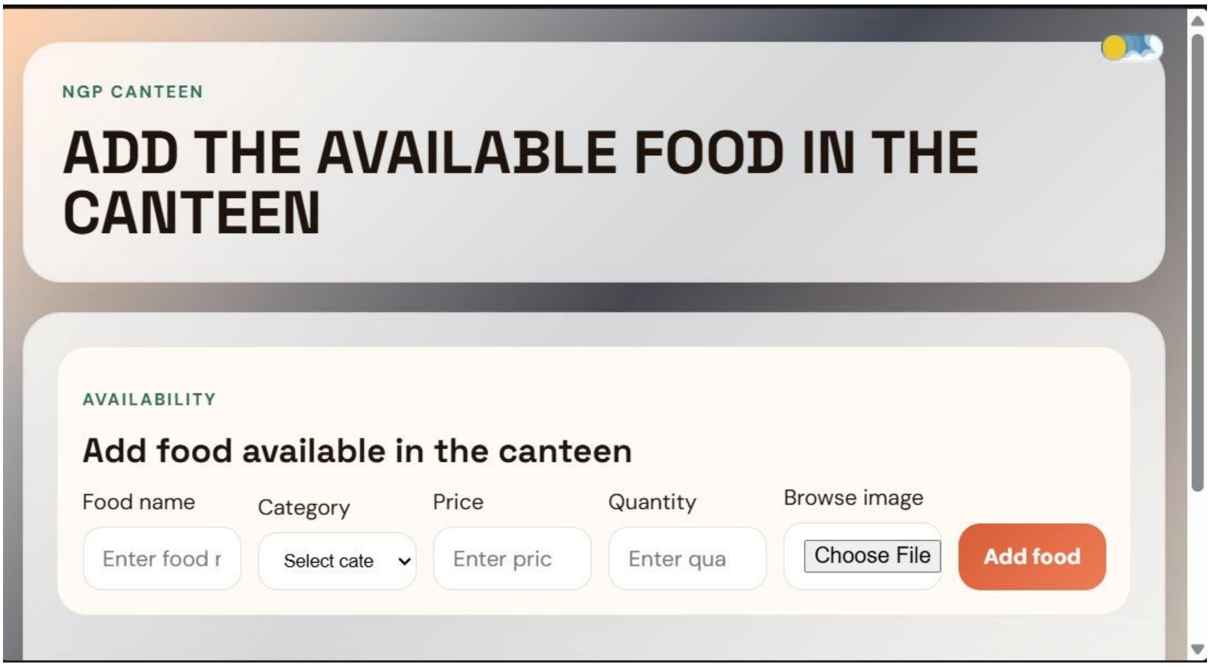
1.Admin Login:

The admin logs into the system using authorized credentials to access the control panel.



2.Menu Management:

The admin can add, update, or remove food items from the menu and set prices accordingly.



3.Viewing Orders:

The admin can view all incoming orders along with token numbers and order details in real-time.

4.Order Processing:

The admin or staff prepares the food based on the received orders in an organized manner.

5.Updating Order Status:

The admin updates the order status (e.g., “Preparing,” “Ready”) in the system, which is reflected on the student’s interface.

6.Token Verification:

At the time of food collection, the admin verifies the digital token shown by the student.

7.Order Completion:

Once the food is handed over, the admin marks the order as completed in the system.

8.Monitoring and Management:

The admin can monitor all orders, manage records, and ensure smooth functioning of the system.

5.4 RESULT

The Digital Food Management System successfully improves the efficiency of the college canteen by reducing waiting time and eliminating long queues. The system ensures smooth order processing and provides real-time updates, enhancing user convenience.

It also helps in reducing food wastage by ensuring that orders are properly tracked and collected on time. The user-friendly interface and digital token system make the entire process organized and efficient.

Overall, the system provides a reliable and effective solution for modernizing canteen operations and improving the experience for both students and staff.

CHAPTER – 6

RESULT AND DICUSSION

6.1 RESULT

The Digital Food Management System successfully enhances the efficiency and organization of college canteen operations by replacing traditional manual processes with a smart digital solution. The implementation of online ordering significantly reduces long queues and waiting time, allowing users to place orders conveniently from anywhere. The token-based collection system ensures a smooth and well-structured method of food distribution, avoiding confusion during peak hours. Real-time order tracking and notification features improve transparency and keep users informed about their order status. The system also minimizes human errors in order processing and billing, leading to more accurate and reliable service. In addition, it helps reduce food wastage by ensuring timely pickup of orders and better management of uncollected items. From the canteen management perspective, the system simplifies handling multiple orders and improves coordination among staff. It also supports better planning through organized data management. Overall, the project successfully achieves its goal of providing a smart, efficient, and user-friendly canteen management system that enhances both user satisfaction and operational performance.

6.2 DISCUSSION

The Digital Food Management System effectively addresses the major challenges of traditional college canteen operations by introducing a smart and organized digital solution. The use of online ordering and a token-based collection system helps in reducing long queues, overcrowding, and waiting time, thereby improving user convenience and satisfaction. Real-time tracking and notification features enhance transparency and ensure that users are well-informed about their order status. From a technical perspective, the system uses web technologies to provide a reliable and user-friendly platform, reducing manual errors and improving efficiency. It also plays a significant role in minimizing food wastage through proper order management and timely collection. Additionally, the system helps canteen staff handle multiple orders efficiently and supports better planning through stored data. Although the system requires internet connectivity and basic technical knowledge, its advantages in improving service quality and operational efficiency make it a valuable solution for modern canteen management.

CHAPTER - 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Digital Food Management System for college canteens provides an effective and modern solution to the limitations of traditional food ordering methods. By introducing online ordering, real-time updates, and a token-based collection system, it significantly reduces waiting time, overcrowding, and manual errors. The system enhances user convenience, improves service efficiency, and ensures better coordination between customers and canteen staff. It also plays an important role in minimizing food wastage through proper order management and timely notifications. Although there are minor challenges such as dependency on internet connectivity and initial setup requirements, the overall benefits outweigh these limitations. The project successfully demonstrates how digital technology can be used to create a smart, organized, and efficient canteen management system.

7.2 FUTURE SCOPE

The Digital Food Management System can be further enhanced by integrating advanced features to improve functionality and user experience. Future developments may include online payment options such as UPI and digital wallets for a fully cashless system. A mobile application can be developed to increase accessibility and convenience for users. The system can also incorporate AI-based demand prediction to help in better menu planning and resource management. Features like personalized recommendations, feedback systems, and rating mechanisms can be added to improve service quality. Integration with IoT devices or smart displays can further automate the process. Additionally, the system can be expanded to multiple canteens or institutions, making it scalable and more versatile. Overall, continuous improvements can make the system more intelligent, efficient, and adaptable to future technological advancements.

APPENDIX – I

SOURCE CODE

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>FeastFlow | Food Ordering</title>
  <link rel="preconnect" href="https://fonts.googleapis.com" />
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
  <link

href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@400;500;70
0&family=DM+Sans:wght@400;500;700&display=swap"
    rel="stylesheet" />
  <link rel="stylesheet" href="styles.css" />
</head>

<body>
  <div class="theme-switch-wrap">
    <label class="theme-switch" for="themeToggle">
      <input class="theme-switch__checkbox" type="checkbox" id="themeToggle" />
      <div class="theme-switch__container">
        <div class="theme-switch__stars-container">
          <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 144 55" fill="none">
            <path fill="currentColor" d="M135.831 3.00688C135.055 3.85027 134.306
4.7422 133.582 5.68108C132.808 4.71872 132.006 3.80264 131.177 2.93552C130.604
2.33595 129.636 2.30053 129.02 2.8568C128.405 3.41306 128.368 4.36498 128.941
4.96454C130.222 6.30515 131.408 7.76065 132.497 9.32086C131.487 10.8055 130.584
12.3938 129.789 14.0687C129.433 14.8181 129.762 15.7072 130.523 16.0543C131.284
16.4014 132.189 16.0751 132.545 15.3257C133.189 13.9698 133.912 12.6755 134.713
11.4473C135.487 12.6996 136.182 14.0213 136.798 15.4051C137.136 16.1642 138.032
16.5114 138.801 16.1805C139.571 15.8496 139.923 14.9659 139.584 14.2068C138.818
12.4865 137.943 10.8512 136.962 9.31811C137.912 7.99222 138.948 6.75175 140.068
5.60286C140.648 5.00838 140.631 4.05602 140.03 3.47625C139.429 2.89648 138.454
2.9124 137.874 3.50687C137.13 4.27015 136.449 5.10295 135.831 6.00062C135.373
5.29282 134.894 4.62766 134.394 4.00688H135.831Z"/>
            <path fill="currentColor" d="M97.9619 9.53906C97.0429 10.4965 96.1669
11.5066 95.3357 12.5672C94.5038 11.5066 93.6277 10.4965 92.7087 9.53906C92.1212
8.92723 91.1415 8.89818 90.5188 9.47418C89.896 10.0502 89.8662 11.0138 90.4537
```

```
11.6256C91.8643 13.0945 93.1101 14.718 94.1908 16.4929C93.1397 18.2235 92.2427
20.076 91.5 22.0503C91.2032 22.8394 91.6174 23.7119 92.425 24.0019C93.2325
24.2919 94.1273 23.8876 94.4241 23.0985C95.0205 21.5131 95.7338 20.0428 96.5638
18.6869C97.4128 20.0619 98.153 21.5296 98.7845 23.0909C99.0907 23.8489 99.9797
24.227 100.751 23.9352C101.523 23.6433 101.911 22.7749 101.605 22.0169C100.866
20.1939 99.9864 18.4887 98.9666 16.9015C100.073 15.2157 101.361 13.6487 102.829
12.2005C103.428 11.6104 103.428 10.6489 102.829 10.0588C102.23 9.46868 101.252
9.46868 100.653 10.0588C99.7449 10.9548 98.9204 11.9478 98.1798 13.0377C97.495
11.9478 96.7543 10.9548 95.9571 10.0588H97.9619Z"/>
      <path fill="currentColor" d="M60.1268 1.21356C59.3465 2.03885 58.5907
2.91045 57.8603 3.82867C57.0972 2.88129 56.3012 1.98389 55.4722 1.13647C54.8943
0.545779 53.924 0.525763 53.3235 1.09177C52.723 1.65778 52.7046 2.60522 53.2825
3.19591C54.5739 4.51906 55.7671 5.9627 56.8621 7.52684C55.8612 8.99351 54.9717
10.5609 54.1936 12.2289C53.8406 12.9854 54.1728 13.8739 54.9359 14.2143C55.699
14.5548 56.6144 14.218 56.9674 13.4615C57.6049 12.0957 58.3203 10.7968 59.1136
9.56472C59.8879 10.8185 60.5838 12.1405 61.2013 13.5307C61.5379 14.2885 62.4348
14.633 63.2044 14.2999C63.9739 13.9669 64.3233 13.0821 63.9867 12.3243C63.2171
10.5909 62.3392 8.94999 61.3529 7.41445C62.3098 6.08338 63.3522 4.8402 64.48
3.68867C65.063 3.09355 65.0476 2.14092 64.4457 1.56169C63.8438 0.982458 62.8771
0.997882 62.2941 1.59301C61.5578 2.34491 60.883 3.17164 60.2697 4.07321C59.7971
3.36295 59.3079 2.6964 58.8019 2.07356H60.1268Z"/>
     </svg>
    </div>
    <div class="theme-switch__clouds"></div>
    <div class="theme-switch__circle-container">
     <div class="theme-switch__sun-moon-container">
      <div class="theme-switch__moon">
       <div class="theme-switch__spot"></div>
       <div class="theme-switch__spot"></div>
       <div class="theme-switch__spot"></div>
      </div>
     </div>
    </div>
   </div>
  </label>
 </div>
 <div class="page-shell">
  <section class="auth-layout" id="loginView">
   <div class="login-container">
    <form class="login-card" id="loginForm">
     <div class="login-title">
      <span class="login-text">NGP Canteen Login</span>
     </div>

     <div class="login-form">
      <div class="input-group">
```

```
          <input class="login-input" type="text" id="username" placeholder="Enter username" required />
        </div>

        <div class="input-group">
          <div class="password-field">
            <input class="login-input" type="password" id="password" placeholder="Enter password" required />
            <button type="button" class="password-toggle" id="passwordToggle" aria-label="Show password">Eye</button>
          </div>
        </div>

        <div class="input-group hidden" id="confirmPasswordGroup">
          <input class="login-input" type="password" id="confirmPassword" placeholder="Confirm password" />
        </div>

        <p class="login-error hidden" id="loginError">Incorrect username or password.</p>
        <p class="signup-success hidden" id="signupSuccess">Account created. Sign in now.</p>
        <button type="submit" class="login-btn">Login</button>
        <button type="button" class="login-mode-btn" id="modeToggleBtn">Create new user</button>
      </div>
    </form>
  </div>
</section>

<section class="app-view hidden" id="appView">
  <header class="hero">
    <div>
      <div class="brand-chip">NGP CANTEEN</div>
      <h1>ADD THE AVAILABLE FOOD IN THE CANTEEN</h1>
    </div>
  </header>

  <main class="content-grid single-column">
    <section class="menu-section">
      <div class="availability-panel">
        <div class="section-head compact">
          <div>
            <p class="eyebrow">Availability</p>
            <h3>Add food available in the canteen</h3>
          </div>
        </div>
```

```html
<form class="add-food-form" id="addFoodForm">
  <label>
    <span>Food name</span>
    <input type="text" id="foodName" placeholder="Enter food name" required
/>
  </label>
  <label>
    <span>Category</span>
    <select id="foodCategory" required>
      <option value="">Select category</option>
      <option value="Breakfast">Breakfast</option>
      <option value="Lunch">Lunch</option>
      <option value="Snacks">Snacks</option>
      <option value="Drinks">Drinks</option>
      <option value="Desserts">Desserts</option>
    </select>
  </label>
  <label>
    <span>Price</span>
    <input type="number" id="foodPrice" placeholder="Enter price" min="1"
required />
  </label>
  <label>
    <span>Quantity</span>
    <input type="number" id="foodQuantity" placeholder="Enter quantity"
min="1" required />
  </label>
  <label>
    <span>Browse image</span>
    <input type="file" id="foodImage" accept="image/*" />
  </label>
  <button type="submit" class="primary-btn" id="foodSubmitBtn">Add
food</button>
</form>
<p class="food-status hidden" id="foodStatus">Food item added to the
canteen menu.</p>
</div>

<div class="section-head">
  <div>
    <p class="eyebrow">Menu</p>
    <h3>Canteen specials</h3>
  </div>
  <div class="pill-row">
    <button type="button" class="pill active" data-category="All">All</button>
    <button type="button" class="pill" data-
category="Breakfast">Breakfast</button>
```

```
        <button type="button" class="pill" data-category="Lunch">Lunch</button>
        <button type="button" class="pill" data-category="Snacks">Snacks</button>
        <button type="button" class="pill" data-category="Drinks">Drinks</button>
       </div>
      </div>

      <div class="menu-grid" id="menuGrid"></div>
     </section>
    </main>
   </section>

   <section class="app-view hidden" id="studentView">
    <header class="hero">
     <div>
      <div class="brand-chip">NGP STUDENT</div>
      <h1>ORDER FOOD FROM THE CANTEEN</h1>
     </div>
    </header>

    <main class="content-grid">
     <section class="menu-section">
      <div class="section-head">
       <div>
        <p class="eyebrow">Order Menu</p>
        <h3>Available foods</h3>
       </div>
      </div>

      <div class="menu-grid" id="studentMenuGrid"></div>
     </section>

     <aside class="cart-panel">
      <div class="section-head compact">
       <div>
        <p class="eyebrow">Cart</p>
        <h3>Your order</h3>
       </div>
      </div>

      <div class="cart-items" id="studentCartItems">
       <p class="empty-state">Your cart is empty. Add food to place an order.</p>
      </div>

      <div class="cart-summary">
       <div>
        <span>Subtotal</span>
```

```html
      <strong id="studentSubtotal">Rs. 0.00</strong>
     </div>
     <div class="total-row">
      <span>Total</span>
      <strong id="studentTotal">Rs. 0.00</strong>
     </div>
    </div>

    <button class="primary-btn checkout-btn" id="studentCheckoutBtn">Place
order</button>
   </aside>
  </main>
 </section>

 <section class="app-view hidden" id="paymentView">
  <header class="hero">
   <div>
    <div class="brand-chip">PAYMENT</div>
    <h1>COMPLETE YOUR PAYMENT</h1>
   </div>
  </header>

  <main class="content-grid single-column">
   <section class="menu-section">
    <div class="availability-panel">
     <div class="section-head compact">
      <div>
       <p class="eyebrow">Payment Summary</p>
       <h3>Review and pay</h3>
      </div>
     </div>

     <div class="cart-items" id="paymentItems"></div>

     <div class="payment-user-panel">
      <label class="payment-user-label" for="paymentUsername">TYPE THE
USER</label>
      <input
       type="text"
       id="paymentUsername"
       class="payment-user-input"
       placeholder="TYPE THE USER NAME"
       autocomplete="name" />
     </div>

     <div class="payment-methods">
      <p class="eyebrow">Method Of Payment</p>
```

```
        <label class="payment-option">
          <input type="radio" name="paymentMethod" value="UPI" />
          <span>UPI</span>
        </label>
      </div>

      <div class="payment-qr-panel">
        <p class="eyebrow">Scan To Pay</p>
        <img
          src="./FOOD ITEM IMAGE/payment-qr.jpeg"
          alt="UPI payment QR code"
          class="payment-qr-image" />
      </div>

      <div class="cart-summary">
        <div>
          <span>Total Amount</span>
          <strong id="paymentTotal">Rs. 0.00</strong>
        </div>
      </div>

      <p class="food-status hidden" id="paymentTokenMessage"></p>

      <div class="payment-actions">
        <button type="button" class="edit-btn" id="backToCartBtn">Back to
cart</button>
        <button type="button" class="primary-btn" id="confirmPaymentBtn">Pay
now</button>
      </div>
     </div>
    </section>
   </main>
  </section>

  <section class="app-view hidden" id="tokenView">
   <header class="hero">
    <div>
     <div class="brand-chip">ORDER TOKEN</div>
     <h1>YOUR ORDER IS CONFIRMED</h1>
    </div>
   </header>

   <main class="content-grid single-column">
    <section class="menu-section">
     <div class="token-panel">
      <p class="eyebrow">Token Number</p>
      <h2 id="orderTokenNumber">NGP0000</h2>
```

```
      <p class="panel-copy">Show this token while collecting your food from the
canteen.</p>

      <div class="cart-items token-items" id="tokenOrderItems"></div>

      <div class="cart-summary">
       <div>
        <span>Total Amount</span>
        <strong id="tokenOrderTotal">Rs. 0.00</strong>
       </div>
      </div>

      <div class="payment-actions">
       <button type="button" class="primary-btn" id="backToMenuBtn">Back to
menu</button>
      </div>
     </div>
    </section>
   </main>
  </section>

  <section class="loader-view hidden" id="loaderView">
   <div class="loader-panel">
    <div class="book" aria-hidden="true">
     <div class="book__pg-shadow"></div>
     <div class="book__pg"></div>
     <div class="book__pg book__pg--2"></div>
     <div class="book__pg book__pg--3"></div>
     <div class="book__pg book__pg--4"></div>
     <div class="book__pg book__pg--5"></div>
    </div>
    <p class="loader-copy">Loading next page...</p>
   </div>
  </section>
 </div>

 <script src="script.js"></script>
</body>

</html>
```